Deforestation Quantification in Satellite Images using Self Supervised Learning & Generative AI

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Introduction

Forests play a vital role in climate regulation, biodiversity, and human well-being, but deforestation driven by logging, agriculture, and pollution threatens ecosystems, with satellite technology and AI now enhancing monitoring and detection of deforestation trends [1, 2, 3, 4].

Satellite image analysis using spectral indices like NDVI and SAVI helps assess vegetation health, traditionally requiring subject-matter expertise (SME) to interpret the indices and environmental factors [5].

Here we investigate how AI can simplify the process by analyzing data patterns and leveraging machine learning to track forest cover changes and quantify deforestation with minimal SME [6, 7].

Methods

We performed object detection through image classification using Multi-Model Feature Fusion powered by Transfer Learning, Generative AI, and Self-Supervised Learning (SSL).

Our approach combines two complementary pre-trained models:

- DINOv3 - A self-supervised vision transformer pre-trained on generic images
- Diffusion model - A generative model with U-Net architecture trained from scratch on labeled satellite and aerial imagery that learns spatial data patterns and textures

Both models remain frozen while we train a lightweight fusion layer that learns how to perform image classification at patch level by combining (fuzing) image features extracted by the above models.

Transfer Learning improves image classification efficiency by leveraging models typically pretrained on large datasets, e.g. over 1.7 billion images sourced from public Instagram posts for DINOv3 [19]. However, a key challenge is that input resolution is standardized to relatively low dimensions (e.g., 224×224 pixels for DINOv3, 128×128 for the custom diffusion model), which may limit the model's ability to capture finer details in high-resolution satellite imagery [11][12].

In applications like deforestation analysis, high-resolution satellite images (e.g., 8192×4320 pixels) are used to capture detailed information about large geographic areas. Traditional image classification models require resizing and cropping, leading to a significant loss of resolution, which can compromise accuracy, especially for tasks requiring fine detail such as estimating forest cover [13]. Our fusion approach aims to mitigate this by combining semantic understanding (DINOv3) with texture-aware features learned from satellite imagery by the diffusion model (U-Net) to perform image patch classification, which is then aggregated to get image segmentation in large satellite images.

DINOv3 Self-Supervised Vision Foundation Model | Meta AI Research, 2025

What is DINOv3? A large-scale **self-supervised image model** that learns general visual understanding from **unlabeled images**, serving as a universal vision backbone.

The Breakthrough

- 7B parameters trained on 1.7B unlabeled images
- First self-supervised model to match/exceed weakly-supervised models
- Learns from raw pixels—no manual labels needed

Gram Anchoring: The Key Innovation

- Maintains patch-level quality throughout training
- Enables ultra-high resolution (up to 4096×4096 pixels)
- Preserves fine-grained spatial details

State-of-the-Art Results (Frozen Backbone)

- 63.0 mIoU – ADE20k segmentation
- 66.1 mAP – COCO object detection
- One model, multiple tasks via lightweight decoders

Why It Matters

Single pretrained model → Many downstream tasks

How Diffusion Models Work

Core Concept:

Diffusion models learn to generate images by reversing a noise-adding process. They gradually transform random noise into structured data through learned denoising steps.

1. Forward Process (Training)

Image → Noise

The scheduler applies noise to images according to a noise schedule over T timesteps (e.g., num_train_timesteps=1000). Each step gradually adds more noise until the image becomes pure random noise.

```
noisy_image = scheduler.add_noise(clean_image,
noise, timestep)
```

2. Reverse Process (Generation)

Noise → Image

During inference, the scheduler generates images from noise. The model iteratively predicts and removes noise at each timestep to recover image structure.

```
noise_pred = model(noisy_image, timestep)

# Iteratively denoise to generate image
```

Training Objective

The model learns to predict the noise that was added at each timestep. Training uses MSE loss between predicted noise and actual noise.

```
loss = MSE(noise_pred, noise)
```

How Diffusion Models Work

Diffusion Model Architecture

Trained from scratch (no pretrained weights) using encoder-decoder structure and UNet2D architecture with skip connections. Includes ResNet blocks and attention layers for spatial self-attention.

Encoder

DownBlock2D and AttnDownBlock2D compress spatial dimensions while increasing channels.

Bottleneck

Lowest spatial resolution with attention layers for semantic understanding.

Decoder

UpBlock2D and AttnUpBlock2D reconstruct spatial dimensions with skip connections.

Typical classification approach for multi-modal diffusion models: Zero-Shot Classification

Pretrained diffusion models can classify images by evaluating epsilon (noise) prediction loss across different text prompts. The class with lowest prediction error is selected.

High-resolution challenge

To address this challenge, we adopted a novel approach that improves quantification accuracy by leveraging image patch classification.

Instead of resizing the entire image, we divided the large satellite images into smaller, non-overlapping patches and applied our classification model to each patch.

The approach benefits from the small patch size required by our model thus functions similarly to an image segmentation model, facilitating detailed forest detection.

By comparing before-and-after satellite images of the same geographic area [14], we can identify deforestation patterns [15] with greater precision, offering actionable insights for environmental conservation efforts.

Implementation

The image classification model was developed using PyTorch, TensorFlow,Keras [16], and transformer libraries. The system combines features from two frozen pre-trained models. DINOv3 was used in a pre-trained state. The diffusion model was trained from scratch for this project using the RESISC45 dataset. The workflow follows two steps.

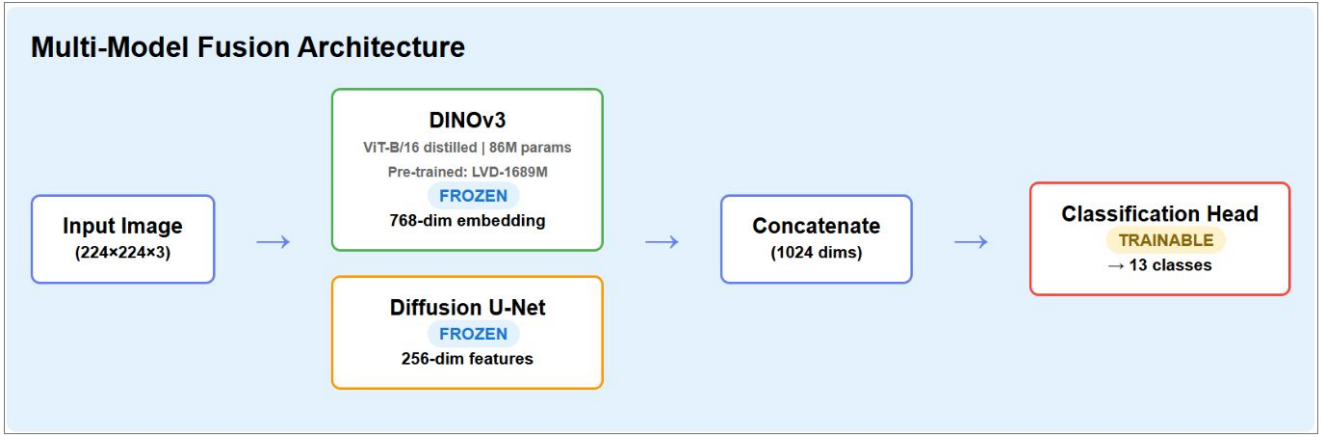
1:Train the diffusion model on the RESISC45 dataset.

2: Extract features from the trained diffusion model [18] and DINOv3 [19], then use these features to train a linear classifier using a standard transfer learning approach.

The linear classifier is implemented as a fusion layer composed of fully connected layers. This layer serves as the only trainable component and leverages the combined embeddings from DINOv3, which produces a 768-dimensional vector, and the intermediate activations from the U-Net diffusion encoder, which produce a 256-dimensional vector.

The model was trained on an NVIDIA GeForce RTX 5080 laptop GPU for about 25 epochs with early stopping, using a batch size of 32 images and the Adam optimizer (learning rate: 1×10^{-4}).

Methods



Model Parameters

Component	Parameters	Status
DINOv3 Encoder	86,021,120	Frozen (Pre-trained)
Diffusion U-Net Encoder	113,641,219	Frozen (Pre-trained)
Fusion + Classification Layers (3)	659,469	Trainable

Total params: 200,321,808 (763.9 MB)
Trainable params: 659,469 (2.5 MB)
Non-trainable params: 199,662,339 (761.4 MB)

Training Details

Parameter	Value
Optimizer	Adam (lr=1e-4)
Batch size	32
Feature extraction	DINOv3: [CLS] token | U-Net: Down block 4 activations

Data: Satellite images dataset used for training

The fusion classification model was trained using the NWPU-RESISC45 dataset, developed by Northwestern Polytechnical University, China [15].

The original dataset includes 45 classes. A subset of 13 large, non-man-made object classes relevant to large-scale forest detection was selected:

'beach' 'circular_farmland' 'cloud' 'desert' 'forest' 'freeway' 'lake' 'meadow' 'mountain' 'rectangular_farmland' 'river' 'snowberg' 'wetland'

The selected subset contained 7,199 images in total.

- training 5,413 images (75%)
- Validation: 1,786 images (25%)

Data: Satellite images used for deforestation quantification

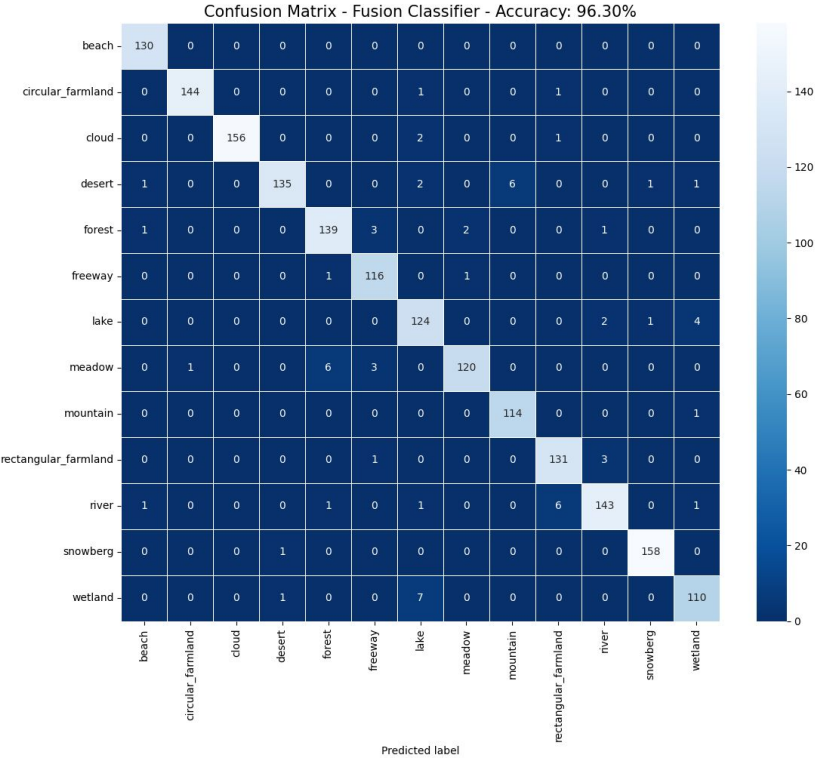
- High-resolution satellite images of deforestation in California's national forests, focusing on the Modoc and Tahoe regions.
- Cover the same geographic areas over time
- Image source: Google Earth [14]

Attribute	Value
Image Size (pixels)	8192 x 4320
Field of View (meters)	6282 x 3189, 5025 x 2550
Pixel Size (meters)	0.75 x 0.75, 0.6 x 0.6

Properties of the manually selected images used for deforestation quantification.

Fused Models Classification Performance

The performance of fused models trained on the subset of RESISC images evaluated on the validation data (not used for training).



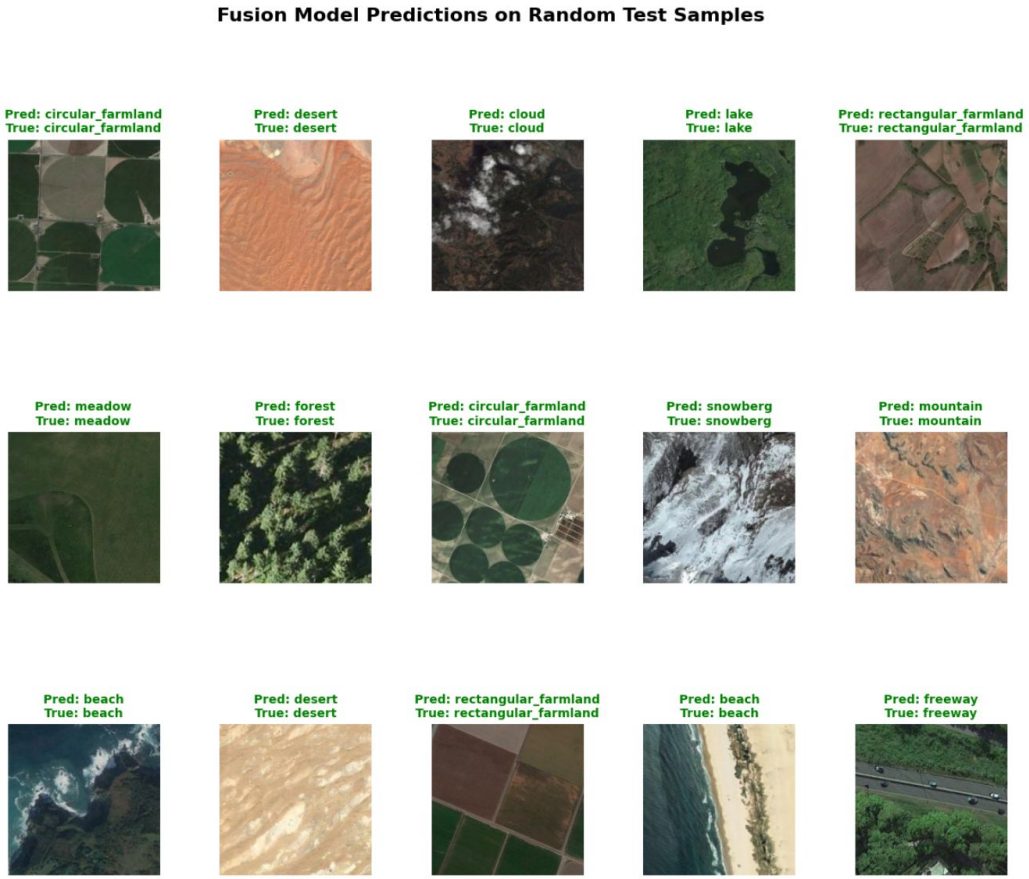
Accuracy and confusion matrix analysis on validation dataset

	precision	recall	f1-score
beach	0.98	1.00	0.99
circular_farmland	0.99	0.99	0.99
cloud	1.00	0.98	0.99
desert	0.99	0.92	0.95
forest	0.95	0.95	0.95
freeway	0.94	0.98	0.96
lake	0.91	0.95	0.93
meadow	0.98	0.92	0.95
mountain	0.95	0.99	0.97
rectangular_farmland	0.94	0.97	0.96
river	0.96	0.93	0.95
snowberg	0.99	0.99	0.99
wetland	0.94	0.93	0.94
accuracy			0.96
macro avg	0.96	0.96	0.96
weighted avg	0.96	0.96	0.96

Model performance at class level

Model predictions on Validation Dataset

Sample predictions



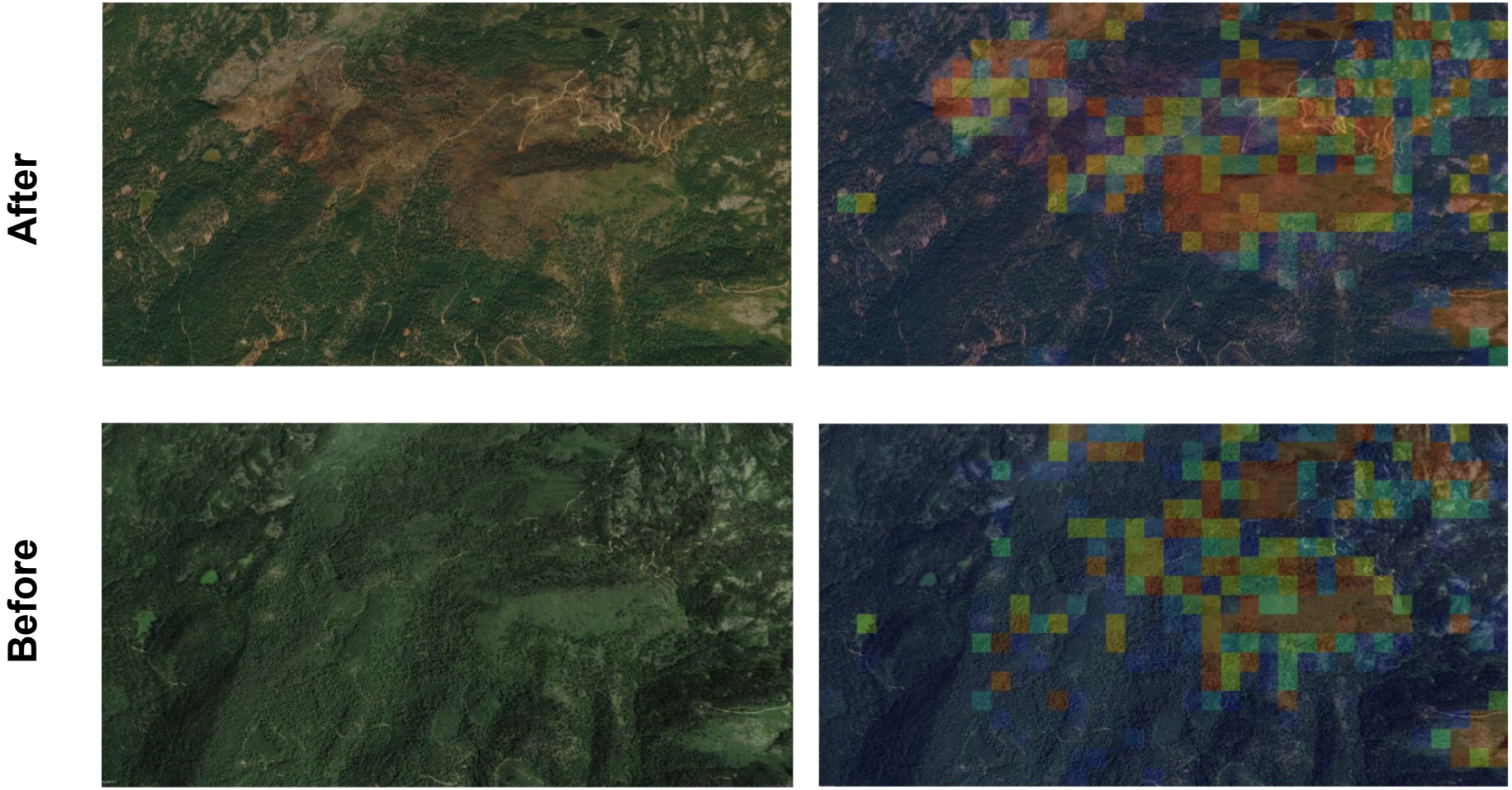
Deforestation quantification

Real-World Scenario

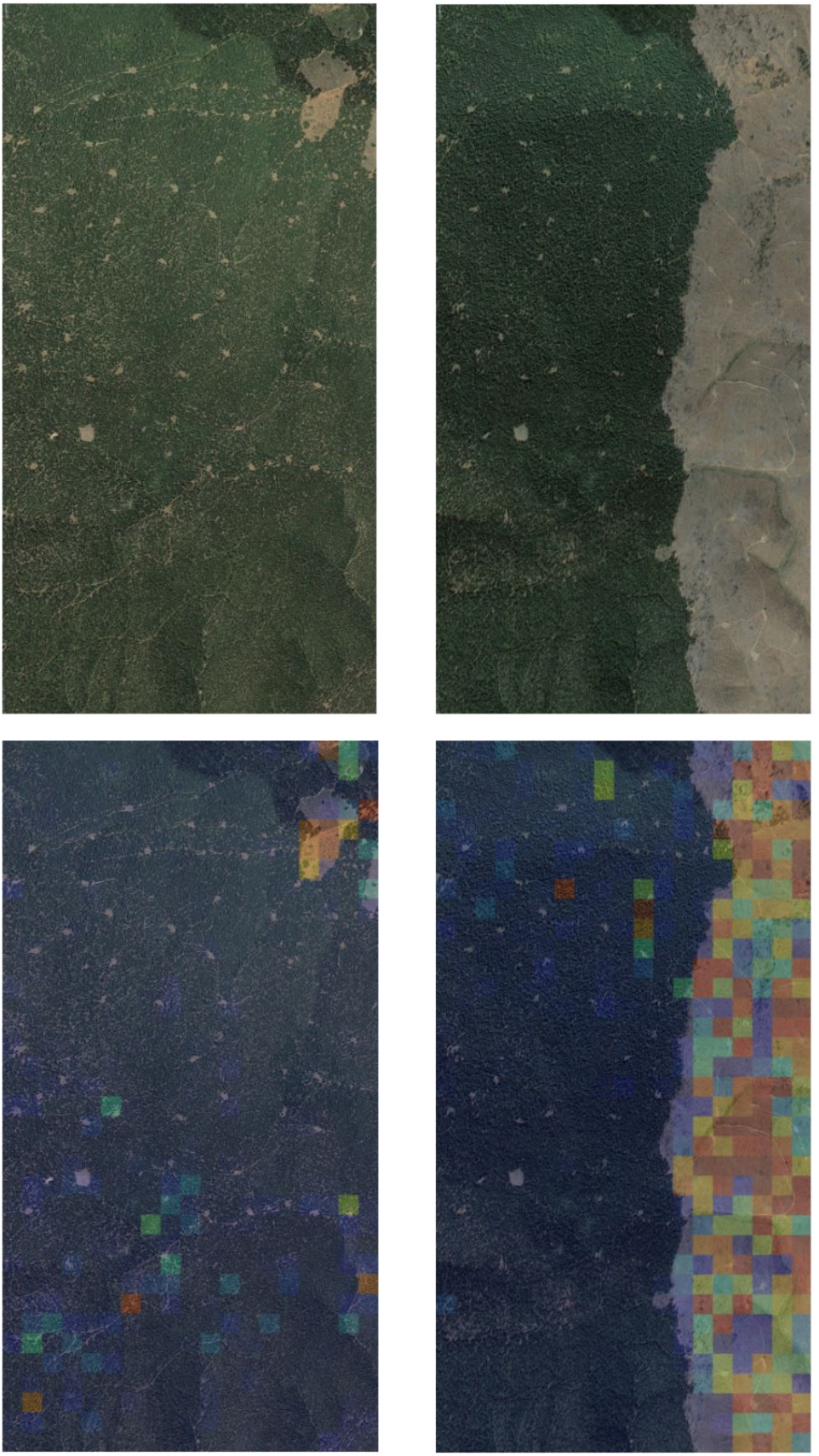
- **Paired Satellite Images**: Two large satellite images were captured, one before a wildfire and one after. These images were divided into smaller patches for analysis.
- **Model Classification**: Each patch was classified individually by the model, predicting the probability of the **Forest** class for each patch.
- **Detailed Detection**: The probabilities were combined to create precise object detection for forest land cover.
- **Heatmap Visualization**: Detection results are displayed as a heatmap, showcasing the spatial distribution of forest cover and deforestation across the satellite images.

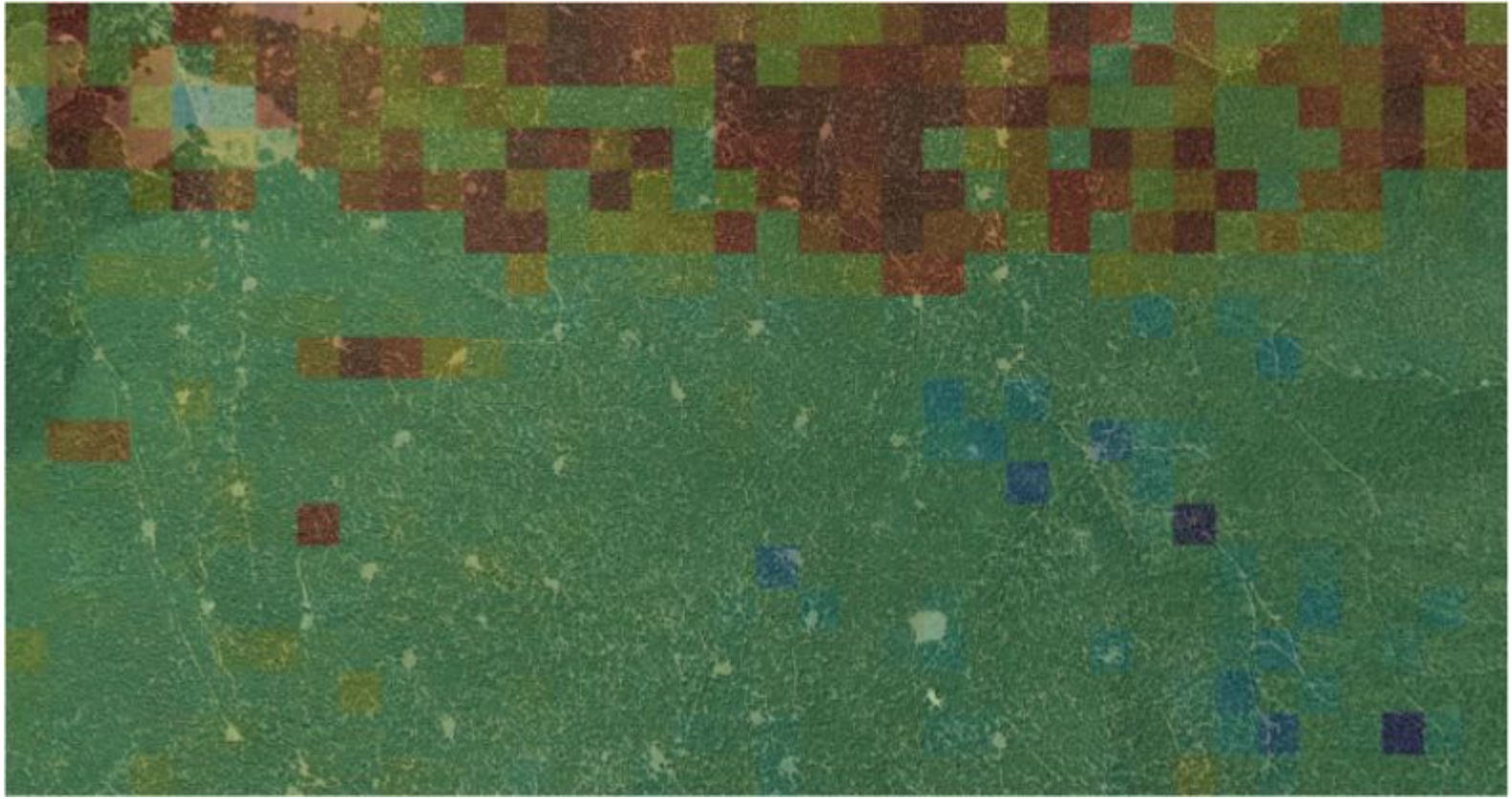
Deforestation Index

- Based on the **Forest class probabilities** predicted by the model.
- Tracks deforestation by excluding areas not covered in both the "before" and "after" images.


After
Before

Before **After**





Deforestation index ignores non forest areas in image before wildfire

Discussion

Our classification model achieved 96.30% accuracy in distinguishing forested from non-forested areas. Its high accuracy on validation dataset makes it a reliable tool for analyzing forest cover changes.

Applied to paired satellite images, the model generated detailed object detection maps of forest land cover, visualized as heatmaps. These provide a detailed detection of forest distribution and deforestation patterns, enabling granular analysis of affected regions.

We developed a Deforestation Index based on Forest class probabilities, focusing on areas with forest cover changes between "before" and "after" images. This approach improves deforestation tracking accuracy and offers a standardized metric for monitoring environmental changes.

Conclusions

Forests play a crucial role in climate regulation, biodiversity, and supporting livelihoods, yet deforestation remains a significant threat. Traditional monitoring techniques are often ineffective due to the vast and remote nature of forests. This project illustrates how deep learning, transfer learning, and AI can be leveraged to accurately quantify deforestation using satellite imagery.

The AI-driven approach allows for scalable, real-time monitoring of environmental changes. By utilizing heatmaps and deforestation indices, stakeholders can effectively visualize and measure forest loss, aiding conservation efforts. This method overcomes challenges such as data scarcity and reduces reliance on subject-matter expertise, highlighting the vital role of AI in detecting illegal logging, tracking habitat changes, and supporting sustainability objectives.

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